

Reparameterisation Invariance Limits the Benefit of Reorganising a Body Code for Morphology Transfer

王牧華, *Department of Biomedical Engineering, National Yang Ming Chiao Tung University, Taiwan. Correspondence: may.be13@nycu.edu.tw. This work received no external funding and the author declares no conflict of interest.*

ABSTRACT

Developmental robotics has long proposed that an agent should master a competence and then reorganise it for a new body. We show that for a large family of such interventions the outcome is settled in advance by an identity. With the dynamics core frozen and only a low-dimensional body code adapted, replacing that code by any invertible linear map of itself and compensating in the weight block the code enters leaves the network's function unchanged, so adaptation begun from the mapped start point under an equivariant update rule cannot change held-out error. The identity needs two conditions, and we measure both on a simulated muscle-driven arm. Function-preserving nonlinear re-encodings cost +0.653 and +0.875 mm ($n = 20$, $p \leq 4 \times 10^{-4}$), while linear ones are bounded above by 0.020 mm and are not resolved from zero. One nonlinear map is the identity at every training code, so it preserves leave-one-body-out decodability by construction, yet it still costs +0.875 mm, which separates decodability from accuracy. Under Adam, equivariant only under permutations, even a rotation displaces error cell by cell. Against a positive control that resolves +1.234 mm, whitening the code is worth +0.025 mm ($p = 0.68$). The code decodes named morphology scalars, though mass decodes only unstably. A fixed master-then-differentiate schedule is worse than joint training at 20 seeds on two of three bodies, under a post-hoc manipulation check. Our nulls are scoped to open-loop prediction on arm2.

Index Terms—Developmental robotics, morphology transfer, negative results, reparameterisation invariance, world models.

I. INTRODUCTION

Developmental cognition holds that agents progressively reformat implicit sensorimotor competence into more explicit, redeployable structure, and developmental robotics has pursued that idea for two decades [1], [2]. Karmiloff-Smith's *Representational Redescription* (RR) [3] frames development as a repeated cycle in which knowledge at first embedded only in successful behaviour is later re-encoded into formats the system can access and recombine, which places that re-encoding *after* the skill has been mastered. The engineering proposal is direct: a model that has mastered one body's dynamics could be redescribed into an explicit, low-dimensional form, and transfer would become a matter of editing a few variables.

We take an engineering operationalisation of that proposal and test it. Since RR proper is endogenous and triggered by the system's own mastery, whereas every mechanism we implement is experimenter-driven, our results bear on the engineering analogue alone, and the null below is neither evidence for nor against RR in humans (Section V).

The substrate is a deterministic, state-based world model of body dynamics, a delta-state predictor $f(s, a) = s + \Delta$ with s the proprioceptive state and a the muscle drive, trained on task-agnostic motor babble for a simulated muscle-driven arm with no pixels and no task reward inside the model. Bodies are dimension-preserving perturbations of a base arm that scale mass and inertia, joint damping and muscle-force gain, so state and action spaces never change and transfer is well-posed.

The mechanisms fall into two families whose asymmetry matters. Mechanisms 1 to 4 test *redescription*, the reformatting of an already-mastered model, whereas mechanism 5 tests *developmental staging*, which is curriculum ordering. The two are developmentally distinct in Karmiloff-Smith's scheme, since RR reformats existing representations where a curriculum reorders experience, so we bundle them only as an engineering master-then-X family and do not treat a staging result as evidence about redescription proper. The distinction is load-bearing, because our one directional result is carried by mechanism 5, the looser analogue, while the mechanism faithful to RR proper that we ran at an interpretable size returned a null.

Terminology. By *body code* (θ) we mean an experimenter-imposed, low-dimensional *input slot* optimised solely to absorb per-body *dynamics* variation, and neither an emergent hidden unit nor a variable the agent owns. We recover only inertial and actuation scalars, namely mass, damping and gain, with no postural, geometric or kinematic content. This is therefore neither the spatial body schema of the neuroscience literature [4], [5] nor the robot body schema of Hoffmann et al.'s review [6], which keeps a representation of where the body's parts are in space. *Emergent* means only that θ is never regressed onto morphology during training; we impose the compression ourselves, and only the code's *alignment* within that slot is emergent.

Contributions. A corollary of two known results, applied to conditioning codes, with its magnitude measured. Neither ingredient is ours, since the GL(n) symmetry of a layer's input block is surveyed in [7] and the optimiser-equivariance taxonomy is textbook [8, §9.5.1], [9]. We add the corollary for a body code, the two conditions under which it fails, namely a moving core or an under-determined fit, and the measurement that the same start-point difference is 0.0016 of a detectable effect under prediction and 2.14 of one under closed-loop control, a 1316-fold difference (Section IV-C4) that scopes our nulls to the prediction metric.

A construction that separates decodability from prediction accuracy. A re-encoding that is the identity at every training code preserves leave-one-body-out decodability by construction, at a difference of 0.000 on all twenty seeds, yet stays nonlinear where adaptation runs and still costs +0.875 mm ([+0.474, +1.275], arm2). Since decoding ground-truth factors is a standard proxy for representation quality, and [10] is a critique of the literature built on it, this is a counterexample to that proxy on the metric we measured.

A calibrated null, pooled over seeds and cells, on the second of the two routes the identity leaves open. Those routes are a nonlinear map and conditioning under a non-

equivariant optimiser. Against a positive control we bound any net benefit from conditioning at 0.099 mm, taken from the interval end most favourable to whitening (Section IV-C2). The control's excursion is larger than the repair whitening performs, however, at condition number 300 against the whitening map's 5.2 at the median seed, so the bound rests on the measured interval.

A test of the master-then-differentiate axis on a learned dynamics world model. Using a fixed schedule and not a competence gate. Staging is *worse* than joint training at 20 seeds on two of three bodies, against a joint arm cut to 1121 optimiser steps versus the staged arm's 1150, though the two are not equalised by learning-rate-weighted budget (Section IV-A, S8).

A factor-specific embodiment-dependence. Damping and gain decode from θ on every body we measured, whereas mass is unstable across seeds, reaching 0.805 on arm2 (12/12 seeds significant) and 0.58 on myoArm (4/4) against medians of -0.16 on arm3 (5/12) and -0.17 on arm4 (4/12). Even the arm3 and arm4 counts are far above the 0.6 of twelve expected by chance ($p = 1.8 \times 10^{-4}$, 2.2×10^{-3}), so what fails is reliability and not decodability. We do not claim monotone degradation with coupling.

Non-claims. We do not claim the first representational redescription in a neural network [11], nor that a decodable context code is novel (CaDM; RMA). We make no claim that the code is available *to the system itself*, and show only that morphology information is linearly recoverable by an external probe, which is strictly weaker. Nor do we claim self/world separation, because the world model is body-only by construction.

II. RELATED WORK

Hidden-parameter and context-conditioned dynamics.

A latent indexing a family of dynamics is the Hidden-Parameter MDP [12], and our body distribution is such a family. What matters here is the distinction between parameter-fed and unsupervised codes, since RMA [13] trains a latent whose *input* is the ground-truth privileged parameters, so decodability back to them is near-by-construction, and PrivilegedDreamer [14] conditions on explicit estimates of them. CaDM [15] infers a context latent from transitions whose latents separate visibly by ground-truth parameter, but it reports no quantitative recovery of *named* physical scalars, and neither PEARL [16] nor the meta-learned adaptive model of Nagabandi et al. [17] recovers named parameters. Our characterisation sits in that gap, since θ is fit unsupervised and never regressed on morphology, yet named log-morphology scalars are linearly decodable from it post hoc. None of these tests a mastery or staging trigger.

Disentanglement and system identification. A deflationary reading is that θ decoding to log-mass, damping and gain is a disentanglement or system-identification result. Locatello et al. [10] prove unsupervised disentanglement *impossible* without inductive biases on both model and data, and classical system identification estimates physical parameters from a running estimator [18]. That literature

offers a candidate account of mass's fragility, since under biarticular coupling a factor can become structurally non-identifiable. Our data do not test it, however, because the most heavily coupled body we ran decodes mass best of the coupled bodies (Section IV-B).

Network-internal redescription, curricula and skill growth. Bartfai's [11] R2MAP implements agent-internal implicit-to-explicit reformatting in an ART network, and is why we do not claim the first redescription in a network; it is stronger than ours in being *triggered* by perceived task difficulty, where our staged arm advances on a fixed schedule. DreamCoder [19] and Voyager [20] grow skill libraries from solved tasks, only Voyager gating on competence, and both add born-explicit skills without reorganising a learned world model. Staged-versus-joint training is an instance of curriculum-versus-i.i.d. training [21], and the automatic-curriculum literature runs from a typology of intrinsic motivation [22] to methods whose benefit is not uniform [23], [24]. Negative curriculum results are therefore not new; what is specific here is a fixed master-then-differentiate schedule applied to a learned dynamics model.

Staged experience in development. The hypothesis is not ours. Thirty years ago Elman [25] showed that a recurrent network could learn a grammar it otherwise failed on when its effective memory was grown on a fixed schedule, though Rohde and Plaut [26] re-ran that setting and the advantage did not survive. Bongard [27] moved the idea from input to body, and his is both the closest precedent to our question and the one clear positive: simulated robots grown across four successive evolutionary stages evolve phototaxis gaits faster and lose less distance under perturbation. That schedule is fixed, as ours is, and not under selection, yet two differences explain why his positive does not predict ours. His schedule acts on an evolutionary search over controllers, where the path a population takes changes what it can reach, whereas ours reorders the data a single fitted model sees, leaving the objective and its optimum unchanged. Moreover his currency is robustness and not held-out prediction accuracy, so a staging benefit there is consistent with a staging cost here. Clark and Karmiloff-Smith [28] supply the representational half, where the currency is again accessibility. All three specify an ordering of experience on a fixed schedule and not a competence gate, which is why our staged arm advances unconditionally.

Parameter-space symmetry and optimiser equivariance.

Two facts we use are established, and we claim neither of them. A layer's input block carries a $GL(n)$ symmetry, surveyed by Zhao et al. [7], and Neyshabur et al. [29] built Path-SGD on a different symmetry but the same underlying point, that gradient descent is not invariant to function-preserving reparameterisations. Whether the symmetry carries from the function to the *estimate* is governed by the optimiser. Newton's method is affine-invariant [8, §9.5.1]; BFGS is not in general, though on this problem we measure L-BFGS to be effectively invariant (Section IV-C1). Adam is equivariant only under permutations, its second moment being an

elementwise square, and we refer to Ling et al. [9] for the rotation failure and to Section IV-C1 for the rescaling failure.

Adjacent lines. Robot body-schema and visual self-modelling systems learn explicit kinematic or dynamic self-models via sensorimotor calibration ([6], [30], [31]), and these represent a self/non-self boundary where ours does not. Controllable-latent world models factorise visual dynamics into controllable and non-controllable parts [32], [33], and symbolic-distillation methods extract closed forms from learned dynamics [34], [35]; both lines are experimenter-run, and neither has a mastery trigger. Finally, fine-tuning has very low intrinsic dimension [36] and LoRA [37] exploits this by confining the update to a low-rank subspace, so a per-body code over a frozen shared core is structurally the same object, which is why the interventions we test have little room left to operate (Section V).

III. METHOD

Bodies. We use planar muscle-driven arms in MuJoCo with 2, 3 and 4 links (arm2, arm3, arm4; state dimensions 8, 14 and 18), plus MyoSuite’s myoArm (20 joints, 34 muscles, 74-dimensional state) and myoHand (23 joints, 39 muscles). Perturbations scale link mass and inertia, joint damping, and muscle-force gain, leaving state and action dimensions unchanged so transfer is well-posed.

World model. The model is a delta-state MLP $f(s, a; \theta) = s + g(\text{norm}(s, a, \theta))$, where g is two hidden layers of 128 SiLU units with input and output normalisers fitted on training data. Training data is task-agnostic motor babble under an Ornstein–Uhlenbeck action process (mean reversion 0.15, noise scale 0.4, actions clipped to $[-1, 1]$, pose re-randomised every 100 steps), 6000 transitions on each of 10 training bodies. The rounds reported here train the core for 25 epochs of Adam at 10^{-3} with batch size 1024. There is no reward and no pixel input anywhere in the model.

Body-code model. A single shared core g is trained jointly over a set of training bodies, each carrying its own free code $\theta \in \mathbb{R}^4$, initialised at $0.1 \times N(0, I)$ and optimised only to reduce prediction error, never regressed onto morphology. The training bodies are the nominal one plus nine log-uniform draws scaling mass by $[0.6, 1.7]$, damping by $[0.4, 3.0]$ and gain by $[0.6, 1.6]$. At test time the core is frozen and only θ is adapted to a held-out body from a small budget of transitions.

The five mechanisms. (1) LPV distillation redescribes the mastered MLP into a local linear-parameter-varying model, against an LPV model fitted directly to the same data. (2) Broad-support distillation repeats (1) over a wider state support generated by the MLP itself, fixing the same-support confound. (3) Few-shot explicit handles adapt only 2-(state-dim) physical knobs, against full fine-tuning. (4) Intervention-based distillation fits the bottleneck by querying the frozen mastered model at chosen inputs, against the same bottleneck fitted on the model’s natural nominal data. (5) Developmental staging versus joint training has a staged learner master the nominal body and then progressively differentiate to new bodies, where a joint learner trains on all bodies at once. Stage advance is an unconditional loop over a fixed epoch count,

with no competence gate anywhere in the code, so we make no mastery-trigger claim. Only mechanisms 4 and 5 carry reportable data; Section IV-A gives the record for the other three and withdraws their numbers.

Metrics. The primary transfer metric is open-loop fingertip prediction error at horizon $k = 6$ on a held-out body after adapting θ , in millimetres, and where a budget sweep is involved an area under the adaptation curve. We additionally report a random-shooting MPC metric and learned-policy SAC+Dyna RL return.

Statistics and pre-registration. Minimum detectable effects are computed as $2.93 \times \text{SD}/\sqrt{n}$ at the round’s own n , where $2.93 = t(.975, 19) + z(.80)$ fixes the 80% criterion at the project’s design size, and projected and achieved values are kept distinct wherever either is quoted. The unit of analysis throughout is the seed, because that is the level at which every manipulation is applied. *A cell that fails its own manipulation check is not interpreted*, including when its unused numbers would have supported us, and Sections IV-A and IV-C each contain an instance. *Not every result below sits under a pre-registration, and we name those that do not*. The five-seed staged-versus-joint comparison in Section IV-A, the staged-versus-joint figures closing Section IV-B, both of Section IV-D’s control metrics and Section IV-B’s twelve-seed decodability result are exploratory. Four of the ten settings in the difficulty round are uncovered, their design document postdating their results; those measurements stand, but the guarantee does not extend to them. S6 gives the conventions and the coverage in full.

Re-encodings of the code. A re-encoding is applied by a wrapper that stores θ in the mapped space and applies the map’s inverse before handing the code to the frozen core, so for corresponding codes the wrapped model computes the same function; Section IV-C1 verifies this across all five re-encodings to 3.8×10^{-6} . Five re-encodings are used, all at $\theta\text{dim} = 4$, and S3 gives them in full. `organised` is the identity, while `k1` and `k300` are linear, $\theta \mapsto \theta A^T$ with log-spaced singular values of geometric mean 1 and condition number $\kappa \in \{1, 300\}$. `scrambled` and `nonlinear_readable` are four-layer additive couplings of the RealNVP form sharing their weights. `nonlinear_readable`, the *gated* map, multiplies each layer’s displacement by a factor that is exactly zero at every training code, so the composition is the identity there and leave-one-body-out $\mathbb{R}^2$ is preserved by construction, while the map stays nonlinear where adaptation runs. The gate width is a tuned hyper-parameter and was not pre-registered, and since the selection rule is monotone by construction and could only ever have returned the smallest candidate, Section IV-C1 reports the cost at all three calibrated widths. The conditioning of a nonlinear map is not a free parameter, so we measure it. The median condition number of the Jacobian, over 24 probe codes per seed, is 90.9 for the gated map (34.2–285.1 across seeds) and 37.7 for `scrambled` (12.0–279.1), against 300 by construction for the linear `k300` control. The linear control is therefore the *worst* conditioned of the three, which is what

makes the comparison interpretable, and whatever the nonlinear maps cost, conditioning is not the explanation.

Adaptation and its start. Few-shot adaptation fits θ only, minimising one-step prediction error on the normalised state delta with no penalty on θ , while the reported metric is a different quantity, open-loop fingertip error after six steps. Two update rules are used, L-BFGS and Adam at three learning rates. Every arm begins from $M\theta_0$, the image of the same point under that arm's own re-encoding, so all arms start from the *same function*, and S5 gives both optimisers' settings. The comparison therefore isolates the geometry the optimiser works in and nothing else, which is what makes the L-BFGS column of Fig. 1 a test of the identity and not of the start. Under Adam the same construction still fails to be free, and that is the point, since satisfying the identity's precondition is not sufficient when the adaptation rule is not equivariant under the map.

IV. RESULTS

A. Staged versus joint training

Two of the five mechanisms carry reportable data. Mechanisms 1–3 were run at two seeds with no stored result file, and the closest stored JSON disagrees with the surviving lab note, so the numbers support no claim and we do not report them. We retain from those three only the *design* contribution, each having fixed a confound that motivated mechanism 4. The five-mechanism framing should therefore be read as two mechanisms with reportable data and three design precursors.

Mechanism 4, intervention-based distillation, is a CI-bearing null whose stored quantity is *direct - redescription*, whose value is -0.224 mm with a 30-cell bootstrap of $[-0.750, +0.319]$ (`results/v2powered_arm2.json`). We report it in the opposite orientation, so a positive number means redescription is worse. Redescription is therefore 0.224 mm worse, seed-level CI $[-0.245, +0.693]$ ($n = 5$, $p = 0.26$), straddling zero and excluding a redescription advantage larger than about 0.25 mm. The 30-cell bootstrap in that orientation is $[-0.319, +0.750]$, at the wrong level: its 30 cells are 6 fixed single-factor perturbations crossed with 5 seeds, so effective body diversity is 6 and it resamples clustered cells, not seeds. Its redescription arm receives *extra* query signal, so the null is conservative.

Mechanism 5, staging versus joint training, is directional, and staging is *worse*. On the best-scaled comparison (arm2, 10 training bodies, 5 seeds, 30 held-out bodies; exploratory, Section III), joint held-out adaptation AUC is 4.44 mm against staged 5.34 mm, a difference of -0.90 mm, with a seed-level interval of $[-1.593, -0.203]$ ($n = 5$, $p = 0.023$). A 30-cell bootstrap would give $[-1.27, -0.55]$, $1.9\times$ narrower and at the wrong level. Staged wins on only 20% (6/30) of held-out bodies, though the marginal per-cell distributions overlap almost entirely (joint $[1.76, 7.09]$ against staged $[2.12, 8.97]$).

The staged and joint arms do not receive equal optimiser steps, since the staged arm takes 78.0% of the joint arm's, or

30.5% once each step is weighted by its learning rate, the gap between the two accountings being the $0.3\times$ core rate at which the differentiation stages run. That reduced rate is the curriculum's operational definition of consolidate-then-differentiate, so equalising the weighted budget would abolish the manipulation; what can be done is to raise it and see whether the direction turns. Under the five conditions `DESIGN_MATCH.md` fixes in advance, the direction survives on both bodies at every budget while the magnitude attenuates on arm3, from -1.238 mm at the 40% weighted budget to -0.811 mm at 69%, a 34% change. At that 40% budget, seventeen of twenty seeds favour joint training on arm2 and eighteen of twenty on arm3. The trend does not reach zero inside the tested range (S8), so the residual step disparity does not plausibly account for the direction, but we cannot say the effect is independent of it.

A third body is excluded, by a criterion that is not the one we pre-registered, and its excluded numbers would have supported us. The pre-registration required θ -adaptation to beat no-adaptation *on every seed*. That clause is defective, because it gets harder as n grows, and at $n = 20$ the best arm on arm2 reaches only 19/20, so applied literally it voids all three bodies, including the two that carry our directional result. We therefore substituted a criterion, the seed-level 95% interval for (adapted – unadapted) lying wholly below zero, and applied it uniformly to all three bodies $\times$ five training conditions. Under it, arm2 and arm3 pass on every arm (mean improvements 1.1 – 3.0 mm, $p \leq 2 \times 10^{-3}$) and arm4 fails on every arm (five contrasts straddling zero, $p = 0.14$ – 0.27), so no arm4 cell may be interpreted. Two disclosures follow, and neither is comfortable. The substitution was made after the per-seed clause was seen to fail at $n = 20$, and the substitute criterion was fixed after the per-body outcomes were visible. It costs us arm4, whose direction agreed with ours, but a reader is entitled to weigh that we changed a pre-registered rule and that the change is what keeps our own result alive.

The two families are asymmetric in a way worth stating. Mechanisms 1–4 are the *closer* analogue to representational-redescription-proper, yet the one with reportable data is a CI-bearing null and the other three were not run at an interpretable size. The directional worse-than-joint result is carried by mechanism 5, which is *curriculum staging*, the looser Karmiloff-Smith analogue.

B. The morphology-aligned body-code

Training a shared core jointly across bodies with a per-body code (θ , dimension 4) yields a code from which named log-morphology scalars are linearly decodable under leave-one-body-out cross-validation. On arm2, over $n = 12$ seeds with no exclusions and a 200-permutation null for each (exploratory, Section III), log mass gives $R^2 = 0.805$ $[0.091, 0.947]$ (12/12 seeds $p < 0.05$), log damping 0.916 $[0.312, 0.976]$ (12/12) and log gain 0.910 $[0.833, 0.969]$ (12/12). Brackets are full per-seed ranges, not intervals, giving the lowest and highest seed.

The non-obvious element is *which factor breaks* (Fig. 2), measured across three embodiments at matched dataset size

(20 bodies $\times$ 12 seeds each). Damping and gain decode strongly on all three (medians 0.92/0.97/0.98 for damping and 0.91/0.94/0.86 for gain, arm2, arm3 and arm4 in that order, with 12/12 seeds significant on every body and factor). Mass is unstable across seeds: the arm2 median is 0.805 with 12/12 seeds significant, against arm3 -0.16 (5/12) and arm4 -0.17 (4/12), individual seeds spanning -0.79 to $+0.73$ (arm3 $[-0.74, +0.73]$, arm4 $[-0.79, +0.69]$).

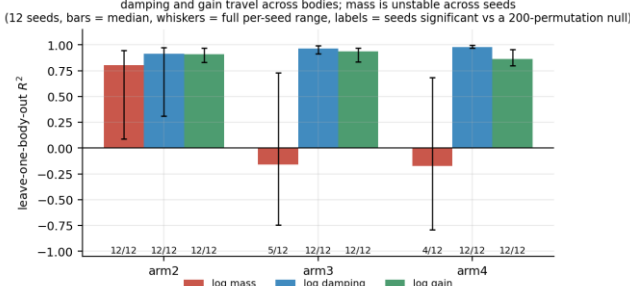


Fig. 2. Leave-one-body-out decodability of named morphology scalars from the unsupervised code, three bodies, 12 seeds. Damping and gain travel everywhere; mass decodes on arm2 but is unstable across seeds on arm3 and arm4, where the median is negative and 5/12 and 4/12 seeds still clear their own null. Labels give the number of seeds significant against a 200-permutation null.

The pattern is not monotone in mechanical coupling. *myoArm*, whose 34 muscles actuating 20 joints make it by far the most coupled body we ran, decodes mass at 0.58 with 4/4 seeds significant, better than either planar coupled arm. What survives is therefore a factor-specific description, weaker than any claim of a trend. On arm3 and arm4 the median is negative, yet 5/12 and 4/12 seeds clear their own permutation null, far above the 0.6 of twelve expected by chance ($p = 1.8 \times 10^{-4}$ and 2.2×10^{-3}), so the supported description is instability across seeds and not failure to decode. A label-shuffle null is negative on all three bodies, and a $\theta_{\text{dim}} = 3$ ablation leaves the arm2 figures essentially unchanged (0.77/0.91/0.90 for mass, damping and gain), so the alignment is neither a cross-validation artifact nor an effect of a slack code dimension. On arm2, joint training aligns better than staged on all three factors (staged 0.46/0.63/0.77 against joint 0.82/0.91/0.90, mass, damping and gain in that order; exploratory, Section III). This does not hold on the other two bodies: on arm3 staged decodes mass better (0.38 against 0.15), and on arm4 staged is ahead on gain (0.87 against 0.85). The contrast does not separate how much body variation the code saw from the order in which it arrived, and all six figures come from 3–4-seed runs, below the 12 seeds used above.

C. Re-encoding invariance and its boundaries

Fix the shared core and adapt only the code, as mechanisms 4 and 5 do and as any method that fits a free context code by gradient descent does. For any invertible M , set $\theta \mapsto M\theta$ and $W\theta \mapsto W\theta M^{-1}$, where $W\theta$ is the single weight block the code enters. Then $f(s, a; \cdot)$ is unchanged for every state, action and body. Adaptation from the mapped start $M\theta_0$ therefore minimises the same objective on the same data, and

if its update rule is equivariant under M the estimate becomes $M\hat{\theta}$, with $f(M\hat{\theta})$ bit-identical to $f(\hat{\theta})$. Under those two conditions held-out error cannot change. Two further conditions hold in our setup, and are stated so a reader can check them elsewhere. The adaptation objective carries no penalty on θ , an L2 penalty not being invariant under M , and θ does not pass through the input normaliser. Any intervention that re-coordinatises the code by an invertible matrix, whether by orthogonalising, rescaling or whitening, is thus inert by construction. Nonlinear re-encodings fall outside the family, and disentangling generally requires one; we measure one below at $+0.875$ mm.

1) The identity and its measurement:

Over 20 seeds and three adaptation budgets under L-BFGS, a random orthogonal reparameterisation costs -0.00007 mm (seed-level CI $[-0.00015, +0.00001]$, $p = 0.075$) and an affine map with condition number 300 costs $+0.006$ mm ($[-0.008, +0.020]$, $p = 0.36$). Both should be read against the construction's own numerical floor, because the five re-encodings agree as functions to 3.8×10^{-6} in the units of the predicted state vector, worth about 1.4×10^{-3} mm of fingertip displacement once propagated through this body's kinematics (median over 200 randomly drawn poses). The rotation map's point estimate is twenty times *smaller* than that. The bound is on how far the arms' functions can differ, not on how precisely a difference can be measured, which is why the rotation map's interval is an order of magnitude narrower than the bound itself. A systematic displacement of 7×10^{-5} mm is therefore fully accounted for by the arms not being exactly bit-identical, so we read it as a bound and not as a cost, and its $p = 0.075$ does not bear on the identity. We apply that reading consistently, so the 0.00017 mm quoted for the rotation below, and the $+0.0001$ mm for whitening in Section IV-C2, sit beneath the same bound. The k_{300} arm at $+0.006$ mm is about four times above it, and the two nonlinear maps two to three orders of magnitude above it. The two function-preserving but nonlinear re-encodings cost $+0.875$ mm ($[+0.474, +1.275]$, $p = 2 \times 10^{-4}$) and $+0.653$ mm ($[+0.333, +0.973]$, $p = 4 \times 10^{-4}$). Linear re-encoding is thus two to three orders of magnitude cheaper than nonlinear (Fig. 1), even though all five maps agree to 3.8×10^{-6} . The identity's linearity condition therefore does real work, and the gated map sharpens the point by preserving the code's leave-one-body-out decodability *exactly* (difference 0.000 on all 20 seeds) and still costs $+0.875$ mm, so decodability is not the operative variable.

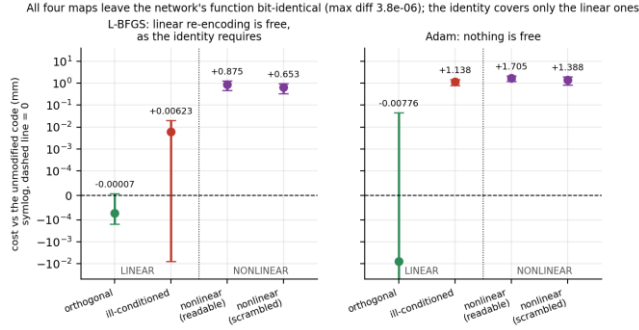


Fig. 1. Linear against nonlinear re-encoding, arm2, 20 seeds, three adaptation budgets pooled, log scale. All four maps leave the network's function unchanged, so what separates them is linearity and not function preservation. Under L-BFGS the linear maps sit at or within four times the construction's numerical floor while the nonlinear maps sit two to three orders of magnitude above it; under Adam even the linear maps stop being free.

Under Adam nothing is free, and that bounds what the identity buys in practice. The ill-conditioned map costs +1.04 to +1.22 mm ($p < 10^{-4}$), the same order as the nonlinear re-encodings. That range is across three learning rates (0.015, 0.05 and 0.15) swept with no pre-registered choice among them, and 0.05 is the middle value on every arm (k300 gives +1.225 / +1.138 / +1.037 at 0.015 / 0.05 / 0.15, in that order), so the sweep is an unregistered freedom we did not exercise in our own favour. The most favourable rate for the nonlinear result is 0.15, where the gated-minus-scrambled contrast reaches +1.272 mm ($p = 0.001$) against +0.317 mm ($p = 0.20$) at the rate we report. L-BFGS takes no learning rate, so the figures above are unaffected. The orthogonal map's *signed* seed-level mean is near zero ($p \geq 0.195$), but that is cancellation and not invariance, since cell by cell it displaces held-out error by 0.036–0.512 mm on average across nine (learning-rate, budget) cells, and up to 4.28 mm in a single cell, against 0.00017 mm under L-BFGS.

Adam's per-coordinate second moment is an elementwise square, which does not commute with a rotation, so Adam is equivariant only under permutations. The rotation failure is Ling et al.'s [9]; the rescaling failure is our own measurement. Plain gradient descent and momentum are orthogonally equivariant, their updates being linear in the gradient, which we verify numerically ourselves. Over 200 random four-parameter least-squares problems we mapped the estimate back and measured $\|M^{-1}\hat{\theta}(M) - \hat{\theta}(I)\|$. Adam's median displacement is 5.0×10^{-2} under an orthogonal map and 7.1×10^{-2} under a diagonal rescaling, against 2.7×10^{-16} for momentum under the orthogonal map. Both rules are zero to machine precision under a permutation (src/verify_equivariance.py, results/equivariance.json). This is a property of the adaptation *path* at a finite budget, not of the optimum, since on this convex problem every displacement falls to machine precision by 1000 steps. What distinguishes the rules at the 80 steps our mechanisms use is that momentum's orthogonal displacement is machine zero at *every* budget,

while Adam's second moment does not commute (results/equivariance_budget.json). The honest scope is therefore this: *re-encodings the adaptation rule treats equivariantly are provably free; others are not, and conditioning becomes a real channel.*

Gate width. We re-ran the whole experiment at the two widths the selection rule could not have chosen. The gated map costs +0.875, +0.878 and +0.733 mm at widths 0.12, 0.18 and 0.25 respectively ($n = 20$, $p \leq 6.5 \times 10^{-4}$), against the ungated +0.653 mm in all three runs. No pair of widths is distinguishable, and the comparison excludes differences larger than about 0.4 mm (S4).

2) Conditioning, against a positive control:

Two routes could still pay: a nonlinear map, which the identity does not cover and which costs +0.875 mm even under L-BFGS, and conditioning under a non-equivariant optimiser. We bounded the second, pooled over seeds and cells, though a benefit concentrated on the worst-conditioned seeds is not excluded (S7). With the core frozen we whitened the code, $\theta \mapsto M\theta$ with $M = \Lambda^{(-1/2)}U^T$ built from the covariance of the training codes alone, since using the held-out body would be reading the test set. Three controls ran on identical seeds, bodies, data, starts and step counts: identity, a random orthogonal map, and an affine map with condition number 300. The round was pre-registered on arm2, 20 seeds $\times$ 4 held-out bodies, budgets {25, 100, 400}, under Adam (lr = 0.05) and L-BFGS. All four manipulation checks passed, since the arms agree to 3.1×10^{-6} , the orthogonal map is null under both optimisers (+0.022 mm, $p = 0.38$ under Adam; -0.00004 mm, $p = 0.13$ under L-BFGS), and the ill-conditioned map is null under L-BFGS (-0.0055 mm, $p = 0.66$) as the identity requires. The fourth is a positive control, and on the same seeds the ill-conditioned map reproduces the known damage under Adam at +1.234 mm, $p = 2.2 \times 10^{-6}$, so the instrument resolves a conditioning effect of about 1.2 mm. Against that instrument, whitening buys nothing (Fig. 3). With the three budgets pooled, $WHITE - ORIG$ is +0.0246 mm under Adam (CI [-0.0986, +0.1479], $p = 0.68$), of the wrong sign and bounded well below what the instrument can see; under L-BFGS it is +0.0001 mm ($p = 0.61$). Because a pooled mean can sit near zero through cancellation, we report this arm cell by cell as well. Under Adam its mean absolute displacement is 0.409, 0.359 and 0.225 mm at budgets 25, 100 and 400 respectively, with a largest single seed-and-body displacement of 3.05 mm, so the pooled mean conceals movement of the same order as the rotation's. What it does not conceal is a benefit, since the signed per-budget means are -0.037, +0.055 and +0.056 mm across those same budgets, whereas the ill-conditioned control displaces 1.410 to 1.627 mm per cell with every budget signed the same way (+1.163 to +1.325). Under L-BFGS the whitened arm moves by at most 0.001 mm. Mean held-out error under Adam is $ORIG$ 3.823, $WHITE$ 3.848, ILL 5.057 and ROT 3.845 mm. This is not a case of there being nothing to fix, as the condition number of the training-code covariance has median 27.1, ranging from 4.1 to 243.7 across seeds. The whitening map that removes it has condition

number 5.2 at the median seed and 15.6 at the worst, against 300 for the control, so the control is calibrated at a larger excursion than the repair performs.

Whether a benefit appears on the badly-conditioned seeds and is diluted by pooling is a question we did not pre-register. We answer it anyway and report the answer whatever it is. The seed-level slope on log condition number is -0.063 mm per log unit (95% CI $[-0.170, +0.044]$, $p = 0.23$), the sign the dilution hypothesis predicts but not resolved, and flat under L-BFGS. This analysis is post-hoc and we label it so (S7, `src/whitening_moderator.py`). This is a bound and not an absence of evidence. On an instrument resolving a $+1.2$ mm conditioning effect, moving the code into its own natural coordinates is worth $+0.02$ mm. Our earlier searches carried no positive control, so their nulls could not separate no effect from an effect the measurement cannot see. We offer it only as an order of magnitude, since the two excursions are not matched, the point estimates differing about fiftyfold while the interval bounds any repair at one-twelfth of the damage. The post-hoc moderator neither establishes nor excludes a larger benefit on the worst-conditioned seeds, and its point estimate extrapolated across the observed range would exceed it. Establishing that reparameterisation damage is asymmetric would need a design that asks the question up front.

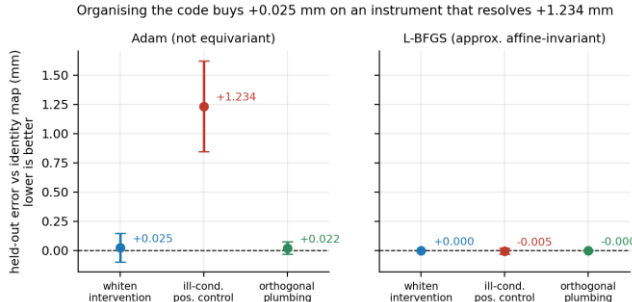


Fig. 3. The conditioning route, tested against a positive control. This is a different experiment from Fig. 1 (20 seeds $\times$ 4 held-out bodies, separately pre-registered), so its ill-conditioned map need not match Section IV-C1's $+0.006$ mm. Whitening the code is worth $+0.025$ mm ($p = 0.68$) on an instrument that the ill-conditioning control shows can resolve $+1.234$ mm ($p = 2.2 \times 10^{-6}$). Under L-BFGS every arm is null, as the identity requires. Bars are seed-level 95% intervals, $n = 20$.

3) Horizon scaling and cross-body scope:

The absolute cost of an ill-conditioned code grows with the evaluation horizon, and the measurement repeats on a second, disjoint seed block. With the adaptation held fixed and only the evaluation horizon varied ($k \in \{6, 20, 50\}$, arm2; Fig. 4), the ill-conditioned map costs $+1.24$, $+7.17$ and $+15.58$ mm at those three horizons in order, a seed-level slope against log k of $+6.67$ (CI $[+3.62, +9.73]$, $p = 2 \times 10^{-4}$). On twenty fresh, disjoint seeds the same measurement gives $+1.91$, $+14.18$ and $+25.01$ mm, slope $+10.86$ (CI $[+6.24, +15.47]$, $p = 1 \times 10^{-4}$), a larger effect than the first block's. Neither block resolves whether the *relative* cost also grows, because as a fraction of baseline the cost is $0.370/0.338/0.308$ on the first block (slope

-0.029 , $p = 0.41$) and $0.533/0.753/0.740$ on the second (slope $+0.102$, $p = 0.18$), both null, but with opposite-signed point estimates and different levels. The first block alone would license the claim that the damage is a roughly constant fraction of baseline error; the second withdraws it.

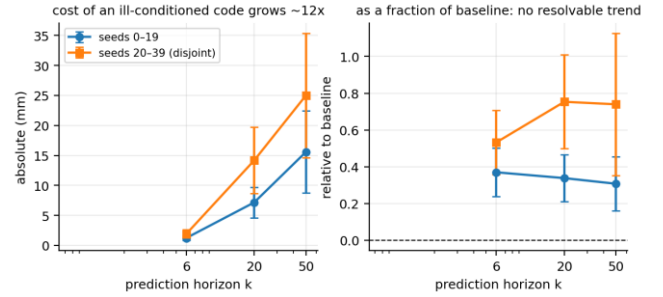


Fig. 4. How the cost of an ill-conditioned code scales with the prediction horizon, on two disjoint seed blocks. Absolute cost grows about thirteenfold; expressed as a fraction of baseline no trend is resolvable in either block, and the two blocks' point estimates carry opposite signs.

A pre-registered entry criterion confines the results above to arm2. A criterion on ten fresh seeds per body, setting no millimetre threshold and asking only whether the instrument resolves a conditioning effect, passes arm2 ($+1.359$ mm, $p = 0.023$) and fails arm3 (-0.007 mm, $p = 0.99$). So arm3 never entered the test, and none of its numbers support a claim; the one quoted here is quoted against us. Since the criterion is defined at $k = 6$ only, its null bounds what we can resolve there and not whether the effect exists. On the same ten seeds the conditioning cost at $k = 50$ is $+32.9$ mm ($p = 0.047$), uncorrected across twelve cells and outside those seeds' authorised use, which is enough to withhold the stronger claim but not to make a result.

4) Where the identity does not apply:

The identity covers re-encoding but not changing f itself, and two conditions would escape it: the core must move, or the fit must be under-determined. Across four bodies and ten settings the spread in held-out error across code initialisations (24 on the three planar bodies, 12 on `myoArm`) never exceeded 3% of that setting's minimum detectable effect projected to $n = 20$, which ranges from 0.221 mm to 1.336 mm; the largest observed ratio was 0.028 . Four of the ten settings carry no pre-registration guarantee (Section III). In-sample fits from different starts agreed to one part in 10^5 – 10^6 , under a curvature-aware optimiser. That scope is load-bearing.

On the same models, data and starts, Adam with a decayed learning rate gives a start-point spread of 1.47 – 1.83 mm at the two smallest adaptation budgets, 10 and 25 transitions, which is 6.7 – $7.2\times$ each setting's own minimum detectable effect (`unconverged_frac` = 0.00, so not a convergence artifact). Our own five mechanisms adapt θ with Adam at $lr = 5 \times 10^{-2}$, 80 steps, and their codebase contains no L-BFGS. The code-fitting problem therefore has one answer under a curvature-aware rule, and we cannot claim our mechanisms sit in the regime where the identity transfers from the function to the estimate. Pushing into the under-determined region was not

measured at a size that settles the idea. Raising the code dimension to 64 or 128 against ten transitions does make start-point spread exceed that effect, but adaptation itself is not demonstrably working there, as the adapted model beats the unadapted one on 2/6 and 3/6 seeds respectively, and across all four under-determined settings the figure spans 0–4 of 6. Those are raw per-seed win counts, which Section IV-A replaces with a seed-level interval wherever a contrast is at stake. We use them only to show the regime is degenerate, and they would not carry a contrast.

A change of dependent variable reaches neither condition, though it does scope the nulls. Every result above uses open-loop prediction error, at six steps except where Section IV-C3 varies the horizon, and a pre-registered test asked whether that choice is doing the work. When the same adapted codes are scored under closed-loop random-shooting MPC, under L-BFGS the spread across eight restarts is 0.001630 of the minimum detectable effect on the prediction metric (1.011 mm) and 2.1441 on the control metric (38.27 mm of MPC cost), both at the twenty seeds the round ran (Fig. 5), an amplification of 1316 $\times$ (arm2, 20 seeds $\times$ 3 held-out bodies $\times$ 8 restarts). It is a ratio of two MDE-normalised spreads and we attach no interval to it, so it is an order-of-magnitude statement about which metric can resolve a start-point difference, not an effect size. It holds even under the pre-registered check guarding against MPC's own sampling noise (35.8 mm against an 82.1 mm effect).

One of the five pre-registered checks was never tested at our unit of analysis, so we re-ran the round at its pre-registered design size. `DESIGN_TASK`'s check 1 required adapted control to beat both the unadapted code and the random-action floor significantly. The four-seed run scored it on point means only and no test was computed; at the seed level its contrasts were -34.0 mm ($p = 0.16$) and -91.2 mm ($p = 0.18$). At twenty seeds the check still cannot be scored, for a reason that is itself the finding: the adapted code is evaluated from eight restarts and the unadapted code from one, so the two are not comparable statistics. The adapted arm's best restart costs 104.1 mm and its worst 188.9 mm, bracketing the unadapted code's 143.3 mm, so on best-of-restarts adaptation wins by -39.2 mm (CI $[-50.2, -28.2]$, $p = 5 \times 10^{-7}$) and on worst-of-restarts loses by $+45.6$ mm (CI $[+32.0, +59.2]$, $p = 1 \times 10^{-6}$). We therefore withdraw check 1 without scoring it, since scoring it would need a restarted control this round did not run. The twenty seeds are 0–19 and include the original four, so this is the design-size completion of that round, not an independent replication.

Two other MPC rounds resolve no adaptation benefit on their pre-registered mean-over-restarts metric. A best-of-restarts comparison is available for the second and for this round, but it is confounded by the restart count in the same way, so we read no direction from it (S2). The round establishes one thing at the twenty seeds it ran (Fig. 5): on a control objective the choice of code start has consequences the prediction metric cannot see. It establishes nothing, however, about whether adaptation itself pays there.

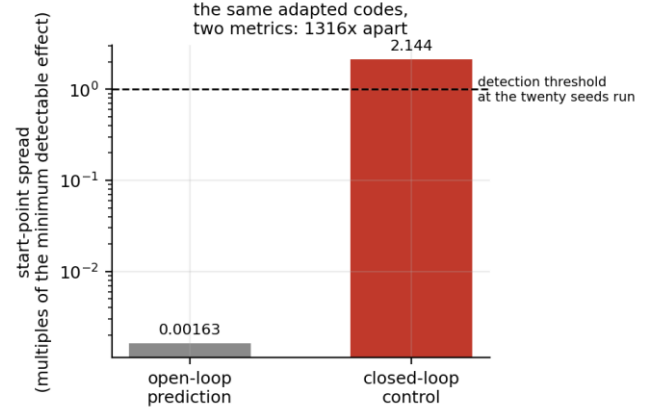


Fig. 5. The same adapted codes scored two ways. Start-point spread is 0.001630 of a minimum detectable effect under open-loop prediction and 2.1441 under closed-loop control, a 1316-fold amplification. Both are normalised by each metric's own minimum detectable effect at the twenty seeds the round ran, so the control bar's crossing of the threshold is a measurement and not a projection.

The change of dependent variable scopes our negative results; it does not overturn them. The identity is untouched, since a function left unchanged returns identical values of *any* downstream measure under an equivariant rule, control objectives included. Only the empirical half narrows, and all our searches for a regime in which organising the code pays used the prediction metric, the calibrated one bounding any net benefit from whitening on arm2 at 0.099 mm. This round used only the untreated arm, by design, so it shows the control metric *can* resolve start-point differences and says nothing about whether organising the code improves them. Our pre-registration names the experiment that would settle it and records that it may not be skipped. Two attempts failed their own pre-registered checks, as the first was underpowered and set no entry condition while the second passed its entry condition and then failed downstream on three counts, one being that adaptation did not beat the unadapted code on the mean-over-restarts metric on a fresh seed block (S2). We add one disclosure the pre-registration did not require. That second round's achieved minimum detectable effect on the test block was 9.05 mm against the 8 mm entry criterion, so applied to the realised run and not the projection, the round would not have opened.

D. Control metrics and compute

Neither control metric supports the joint-over-staged ranking once analysed at the seed level, and both are exploratory (Section III). Random-shooting MPC makes staged 27.7 mm worse, with a seed-level interval of $[-33.7, +89.2]$ ($n = 4$, $p = 0.25$; one of four seeds reverses). Learned-policy SAC+Dyna gives joint – staged return-AUC $+0.10$, $[-0.06, +0.26]$ ($n = 3$, $p = 0.11$). The world model does beat model-free SAC ($+0.99$, $[+0.47, +1.51]$, $p = 0.015$), but only at $n = 3$ seeds, at or below the sizes at which we declined to interpret the contrasts above, so we report it and do not lean on it. The staging conclusion therefore rests on the prediction

proxy alone, and we can no longer claim corroboration outside it.

The whole pipeline runs in minutes of CPU on a single consumer laptop in pure MuJoCo, with the pilot screen completing in 43 s. Every experiment above is re-runnable at that cost.

V. DISCUSSION

The part of this work that generalises beyond our arms is not a measurement. When a shared dynamics core is frozen and only a low-dimensional code is adapted under a rule equivariant to the map, re-coordinatising that code by an invertible matrix is inert by construction, so orthogonalising, rescaling and whitening are all excluded before an experiment is run. What makes that useful rather than trivial is that the conditions the argument needs are the ones a practitioner most often violates without noticing. Nonlinear re-encodings fall outside the family, and disentangling generally asks for one; and the identity carries from the function to the estimate only under an equivariant adaptation rule, which Adam is not.

Those two conditions leave two routes by which reorganising a code might still pay, and we put a calibrated bound on one. Against a positive control demonstrated on the same seeds to resolve a +1.234 mm conditioning effect, whitening the code is worth +0.025 mm on net, and its per-cell displacement is symmetric where the control's is systematic (Section IV-C2). Both figures are arm2, however, and the one attempt to show the same effect on a second body did not resolve it (Section IV-C3). That is a bounded statement and not an absence of evidence, which is why a reader need not take the null on trust.

The other route, nonlinearity, we measured but did not close. The gated map is what we would put to anyone reading reorganisation as disentanglement. Built to be the identity at every training code, it preserves leave-one-body-out decodability to machine precision and still costs +0.875 mm. Keeping a code exactly as readable does not keep it equally useful, so a proxy that many representation-learning arguments rest on comes apart, on our metric, from the property it stands in for.

The empirical half belongs to an older tradition than the algebra, since our one directional result sits with the staging tradition Section II sets out: Elman [25], its failed replication by Rohde and Plaut [26], and Bongard's [27] move from input to body. All of them specify an ordering of experience, and so does ours. There is no competence gate anywhere in our code, so the result speaks to that tradition and not to competence-triggered development, and read against Rohde and Plaut in particular, a staging cost is not an anomaly. That lineage also fixes what the null does not license. What fails is the *engineering* hypothesis, and this is not evidence for or against RR as an account of human development, since a null on a simulated arm cannot bear on such a theory. The point is sharpest on Karmiloff-Smith's E1 level, at which knowledge is redescribed into a format available *as data to the system itself*. Our decodability shows only that morphology information is linearly recoverable by an external probe, so the available-to-

the-agent criterion is not met. Nor did the theory promise what the engineering hope wants of it, because RR is paid in accessibility and not accuracy.

A deflationary reading of the positive half is that θ is a disentangled system-identification code of the same character as RMA's extrinsics latent. We accept it, since discovering such a code is not our contribution; what the developmental framing adds is the staging test and the characterisation of which factors the code isolates. One bridge is speculative and we mark its limit, since a morphology code developed by a forward model might be read as the *dynamics* analogue of a body schema, though the analogy is by functional role only and it should not be cited as an emergent body schema.

If reorganisation does not pay here, it is worth asking where it does. Fine-tuning of large pre-trained models sits in the under-determined regime, and there a low-rank constraint on the adaptation space matches or beats full fine-tuning [37]. That structure constrains the adaptation space and does not better encode an existing code, and our body codes are already at that endpoint, a four-dimensional θ against a frozen core being such a subspace. Starting at the destination is therefore the most economical account of why the journey buys nothing here, and it points to a recommendation we did not test, to seek structure as a constraint on an over-parameterised adaptation space. The same reasoning indicates where a payoff would be visible if one exists. If redescription buys accessibility, it will show up where recombination is required, in novel goals composed from mastered parts, in reuse of a body model across tasks it was not fit for, and wherever the code must be manipulated and not re-estimated. We did not test that, and this is a post-hoc reframing of a pre-registered null. We record it because the alternative reading, that our null counts against RR, is unsupported. For anyone building such systems the practical residue is short: expose the world model to a distribution of bodies, and do not expect a fixed reordering of that experience to pay on a prediction metric.

VI. LIMITATIONS

Bodies. The main experiments use 2-, 3- and 4-link planar arms; 3D morphologies are untested. Re-running mechanism 5 unchanged on *myoArm* (20 joints, 34 muscles) returns a null (joint – staged = +0.19 mm, seed-level CI [–24.08, +24.47]) whose minimum detectable effect at the 4 seeds $\times$ 6 held-out bodies actually run is 22.4 mm (S1), against the 0.90 mm effect measured on arm2, so it is uninformative and not disconfirming. The shortfall is structural and not budgetary, because what remains after the arms share initialisation and training data is the divergence of two optimisation trajectories, 84% of the between-seed variance, and pairing cannot remove it, since the trajectories diverge by seeing different data in a different order, which is the treatment itself. Adding bodies barely helps, as 6 $\rightarrow$ 24 bodies moves the MDE only from 22.4 mm to 21.0 mm at four seeds, and 50 seeds $\times$ 24 bodies would still leave 5.9 mm. Reaching 0.90 mm would need of order two thousand seeds or a lower-variance training procedure. The decodability result does extend to that body

(all three factors significant on 4/4 seeds), so the limitation is specific to the staging result.

Reproducibility of the mechanism-5 figures. The adaptation start point is drawn from a global RNG and not from the reported seed, so the absolute AUCs of mechanism 5 are not reproducible from the published seed, and neither is the paired contrast at the precision we print. Restoring the call order moves the gap from -0.90 mm to -1.282 mm at the same $n = 5$, and to -1.374 mm at $n = 20$, a 42–53% change. That -1.374 mm is the *unmatched* contrast the original run made. Direction and significance survive and the re-measured effect is larger (seed-level CI at $n = 20$ $[-1.843, -0.905]$), so no conclusion turns on it, but -0.90 mm should be read as reproducible only to within about a factor of 1.5.

Task family and dependent variable. All bodies are single-arm reaching systems and the primary metric is open-loop prediction error. The two control metrics do not corroborate the staging ranking at seed level (Section IV-D), and a pre-registered test shows the choice of metric is not neutral, since start-point differences worth 0.001630 of a minimum detectable effect under prediction are worth 2.1441 under closed-loop control (Section IV-C4). The negative results are therefore scoped to the prediction metric. The experiment that would settle whether organising the code pays under a control objective is named in our pre-registration as a step that may not be skipped, both attempts failed their own checks (S2), and this is the most consequential piece of unfinished work here.

The residuals-per-unknown regularity. This regularity is not a law, and we record it as an observation absent from Section IV. Across 14 settings, the influence of start-point choice tracks the number of residuals in the code fit per free code parameter (Spearman -0.923 , $p = 2.6 \times 10^{-6}$), but four settings failed the project's own manipulation check, and restricted to the ten interpretable ones the correlation weakens to -0.811 ($p = 4.4 \times 10^{-3}$). Of those ten, a further four are settings whose pre-registration guarantee Section III withdraws. A pre-registered out-of-sample test on *myoHand*, with predictions and a ± 2 SD band fixed before the run, placed three of four new settings outside the band.

Adaptation optimiser. All five mechanisms adapt θ with Adam ($\text{lr} = 5 \times 10^{-2}$, 80 steps); their codebase contains no L-BFGS, and the L-BFGS measurements of Section IV-C come from the separate follow-up codebase. The identity transfers from the function to the estimate only for maps the adaptation rule is equivariant under, and Adam's class is the permutations alone. The start-point stability that closes the under-determination escape is likewise an L-BFGS result, since under Adam the same spreads are $6.7\text{--}7.2\times$ their own minimum detectable effects rather than under 3% of them. We therefore cannot claim these mechanisms sit in the regime where the identity binds their estimates, only that it binds the function they compute.

Non-claims. We claim no mastery trigger, no monotone relation between mass decodability and mechanical coupling, no general ordering of joint over staged decodability, and no

control-metric corroboration of the staging result. Four results survive, each resting on more than a pilot: the invariance argument, whose magnitude is measured on one twenty-seed round; the staging-worse result at 20 seeds on two bodies, under a manipulation-check criterion substituted post hoc (Section IV-A); the factor-specific decodability characterisation at 12 seeds, which is exploratory; and the calibrated conditioning null on arm2.

VII. CONCLUSION

Training a body-dynamics world model across a distribution of bodies yields a low-dimensional unsupervised code whose alignment to true morphology we characterise under leave-one-body-out cross-validation, and whose decodability is *factor-specific*: damping and gain travel across every body we measured, while mass is unstable across seeds, not absent. Our modest positive is that an unsupervised code of this kind nonetheless decodes named physical scalars. The developmental lever one would expect, master-then-differentiate redescription or staging, is not what produces or exploits it. Neither mechanism carrying reportable data shows the hoped-for benefit, the redescription one being a null and the staging one a reversal, with staging *worse* than joint training at 20 seeds on two of three bodies and the joint arm's step count cut to 1121 against the staged arm's 1150 (Section IV-A, S8).

Two things make that stronger than a bare null. A large sub-family of the interventions the hope reaches for is excluded by an identity, not by measurement, since with the core frozen, re-encoding the code cannot change held-out error under an equivariant adaptation rule. Moreover the second of the identity's two loopholes, conditioning under a non-equivariant optimiser, was tested against a positive control that worked, and whitening moves held-out error by $+0.025$ mm on net ($p = 0.68$), the wrong sign, while degrading its conditioning costs $+1.234$ mm ($p = 2.2 \times 10^{-6}$). Section IV-C3 reports a round whose own control failed on a second body, which is why we keep the arm2 result and the arm3 non-result separate.

This does not license the claim that redescription fails, because the mechanism Karmiloff-Smith describes is not paid in the currency our transfer metric measures (Section V). It licenses something narrower, in that the *engineering* hope attaching a performance currency to that mechanism is not borne out on our proxy, and what the code does carry, linear decodability by an external probe, arrives in a model that has simply experienced body variation.

DATA, CODE AND PRE-REGISTRATION AVAILABILITY

Simulator, training code, analysis scripts, every pre-registration document, and the stored result files behind every number in this paper are available at <https://github.com/maurice1128/body-codes>. Each pre-registration fixes the decision rule and the manipulation checks in advance, and twenty-one of the twenty-four also contain the conclusion paragraph for every outcome written

verbatim. `PREREG_HASHES.md` archives design-document fingerprints and, for those that existed when it was opened, creation times, while `README_PREREGISTRATIONS.md` indexes every round in English, the design documents themselves being in Chinese. Where that record does not reach, at the settings named in Section III and sixteen result files from the exploratory rounds, the repository's `CORRECTIONS.md` gives the itemised account.

ACKNOWLEDGEMENTS

Generative AI (Claude, Anthropic [38]) was used to assist with implementation of experiment and analysis code, and with drafting and editing the manuscript text, including the text of every section of this article. It was not used to generate data, results, or references. All experimental designs, pre-registered decision rules and manipulation checks, statistical choices, and the interpretation of every result were determined by the author, who takes full responsibility for the content. Every numerical claim in this paper was re-derived from the stored raw result files by verification scripts included in the repository.

REFERENCES

- [1] M. Lungarella, G. Metta, R. Pfeifer, G. Sandini, "Developmental robotics: a survey," *Connection Science*, 15(4), 151–190, 2003.
- [2] M. Asada, K. Hosoda, Y. Kuniyoshi, H. Ishiguro, T. Inui, Y. Yoshikawa, M. Ogino, C. Yoshida, "Cognitive Developmental Robotics: A Survey," *IEEE Transactions on Autonomous Mental Development*, 1(1), 12–34, 2009.
- [3] A. Karmiloff-Smith, *Beyond Modularity: A Developmental Perspective on Cognitive Science*. Cambridge, MA, USA: MIT Press, 1992.
- [4] H. Head, G. Holmes, "Sensory disturbances from cerebral lesions," *Brain*, 34(2–3), 102–254, 1911.
- [5] S. Gallagher, *How the Body Shapes the Mind*. Oxford, U.K.: Oxford Univ. Press, 2005.
- [6] M. Hoffmann, H. Marques, A. Arieta, H. Sumioka, M. Lungarella, R. Pfeifer, "Body Schema in Robotics: A Review," *IEEE Transactions on Autonomous Mental Development*, 2(4), 304–324, 2010.
- [7] B. Zhao, R. Walters, R. Yu, "Symmetry in Neural Network Parameter Spaces," *arXiv:2506.13018*, 2025.
- [8] S. Boyd and L. Vandenberghe, *Convex Optimization*. Cambridge, U.K.: Cambridge Univ. Press, 2004, §9.5.1.
- [9] S. Ling, N. Sharp, A. Jacobson, "VectorAdam for Rotation Equivariant Geometry Optimization," *NeurIPS* 2022.
- [10] F. Locatello, S. Bauer, M. Lucic, G. Rätsch, S. Gelly, B. Schölkopf, O. Bachem, "Challenging Common Assumptions in the Unsupervised Learning of Disentangled Representations," *ICML* 2019.
- [11] G. Bartfai, "An Adaptive Resonance Theory-based Neural Network Capable of Learning via Representational Redescription," *Proc. IEEE International Joint Conference on Neural Networks (IJCNN'98)*, Anchorage, 1998.
- [12] F. Doshi-Velez, G. Konidaris, "Hidden Parameter Markov Decision Processes: A Semiparametric Regression Approach for Discovering Latent Task Parametrizations," *IJCAI* 2016, pp. 1432–1440.
- [13] A. Kumar, Z. Fu, D. Pathak, J. Malik, "RMA: Rapid Motor Adaptation for Legged Robots," *RSS* 2021.
- [14] M. Byrd, J. Crandell, M. Das, J. Inman, R. Wright, S. Ha, "PrivilegedDreamer: Explicit Imagination of Privileged Information for Rapid Adaptation of Learned Policies," *ICRA* 2025.
- [15] K. Lee, Y. Seo, S. Lee, H. Lee, J. Shin, "Context-aware Dynamics Model for Generalization in Model-Based Reinforcement Learning (CaDM)," *ICML* 2020.
- [16] K. Rakelly, A. Zhou, C. Finn, S. Levine, D. Quillen, "Efficient Off-Policy Meta-Reinforcement Learning via Probabilistic Context Variables (PEARL)," *ICML* 2019.
- [17] A. Nagabandi, I. Clavera, S. Liu, R. S. Fearing, P. Abbeel, S. Levine, C. Finn, "Learning to Adapt in Dynamic, Real-World Environments through Meta-Reinforcement Learning," *ICLR* 2019.
- [18] L. Ljung, *System Identification: Theory for the User*, 2nd ed. Englewood Cliffs, NJ, USA: Prentice-Hall, 1999.
- [19] K. Ellis, C. Wong, M. Nye, M. Sablé-Meyer, L. Morales, L. Hewitt, L. Cary, A. Solar-Lezama, J. B. Tenenbaum, "DreamCoder: bootstrapping inductive program synthesis with wake-sleep library learning," *PLDI* 2021.
- [20] G. Wang, Y. Xie, Y. Jiang, A. Mandlkar, C. Xiao, Y. Zhu, L. Fan, A. Anandkumar, "Voyager: An Open-Ended Embodied Agent with Large Language Models," *arXiv:2305.16291*, 2023.
- [21] Y. Bengio, J. Louradour, R. Collobert, J. Weston, "Curriculum Learning," *ICML* 2009, pp. 41–48.
- [22] P.-Y. Oudeyer, F. Kaplan, "What is intrinsic motivation? A typology of computational approaches," *Frontiers in Neurobotics*, 1(6), 2007.
- [23] A. Graves, M. G. Bellemare, J. Menick, R. Munos, K. Kavukcuoglu, "Automated Curriculum Learning for Neural Networks," *ICML* 2017.
- [24] R. Portelas, C. Colas, K. Hofmann, P.-Y. Oudeyer, "Teacher algorithms for curriculum learning of Deep RL in continuously parameterized environments (ALP-GMM)," *CoRL* 2019.
- [25] J. L. Elman, "Learning and development in neural networks: The importance of starting small," *Cognition*, 48(1), 71–99, 1993.
- [26] D. L. T. Rohde and D. C. Plaut, "Language acquisition in the absence of explicit negative evidence: How important is starting small?," *Cognition*, vol. 72, no. 1, pp. 67–109, 1999.
- [27] J. Bongard, "Morphological change in machines accelerates the evolution of robust behavior," *PNAS*, 108(4), 1234–1239, 2011.
- [28] A. Clark, A. Karmiloff-Smith, "The cognizer's innards: A psychological and philosophical perspective on the development of thought," *Mind & Language*, 8(4), 487–519, 1993.

- [29] B. Neyshabur, R. Salakhutdinov, N. Srebro, “Path-SGD: Path-Normalized Optimization in Deep Neural Networks,” *NeurIPS* 2015.
- [30] M. Hoffmann, “Body models in humans, animals, and robots: Mechanisms and plasticity,” in *Body Schema and Body Image: New Directions*, Y. Ataria, S. Tanaka, and S. Gallagher, Eds. Oxford, U.K.: Oxford Univ. Press, 2021, pp. 152–180.
- [31] R. Kwiatkowski, H. Lipson, “Task-agnostic self-modeling machines,” *Science Robotics*, 4(26), eaau9354, 2019.
- [32] M. Pan, X. Zhu, Y. Wang, X. Yang, “Iso-Dream: Isolating and Leveraging Noncontrollable Visual Dynamics in World Models,” *NeurIPS* 2022.
- [33] M. Serra Nunes, A. Dehban, Y. Demiris, J. Santos-Victor, “Ego-Foresight: Self-supervised Learning of Agent-Aware Representations for Improved RL,” *arXiv:2407.01570*, 2024.
- [34] M. Cranmer, A. Sanchez-Gonzalez, P. Battaglia, R. Xu, K. Cranmer, D. Spergel, S. Ho, “Discovering symbolic models from deep learning with inductive biases,” *NeurIPS* 2020.
- [35] N. Zolman, C. Lagemann, U. Fasel, J. N. Kutz, S. L. Brunton, “SINDy-RL: Interpretable and Efficient Model-Based Reinforcement Learning,” *Nature Communications*, 16, 10714. (arXiv:2403.09110), 2025.
- [36] A. Aghajanyan, S. Gupta, L. Zettlemoyer, “Intrinsic Dimensionality Explains the Effectiveness of Language Model Fine-Tuning,” *Proc. ACL-IJCNLP 2021*, 7319–7328, 2021.
- [37] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, W. Chen, “LoRA: Low-Rank Adaptation of Large Language Models,” *arXiv:2106.09685 (ICLR 2022)*, 2021.
- [38] Anthropic, “Claude,” large language model. [Online]. Available: <https://www.anthropic.com/claude>